

5-2 解: (a)

$$X = \bar{A} + \bar{A} \cdot B + A \cdot C$$

$$= \bar{A} + C + \bar{A} \cdot B$$

$$= \bar{A} + C$$

(b)

$$X = \bar{A} \cdot B + \bar{A} \cdot C \cdot D + D \cdot B \cdot \bar{D}$$

5-6 解: (a) $X = (A+B) \cdot (C+D)$

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(b)

$$X = \bar{A} \cdot B \cdot C + \bar{C} \cdot D$$

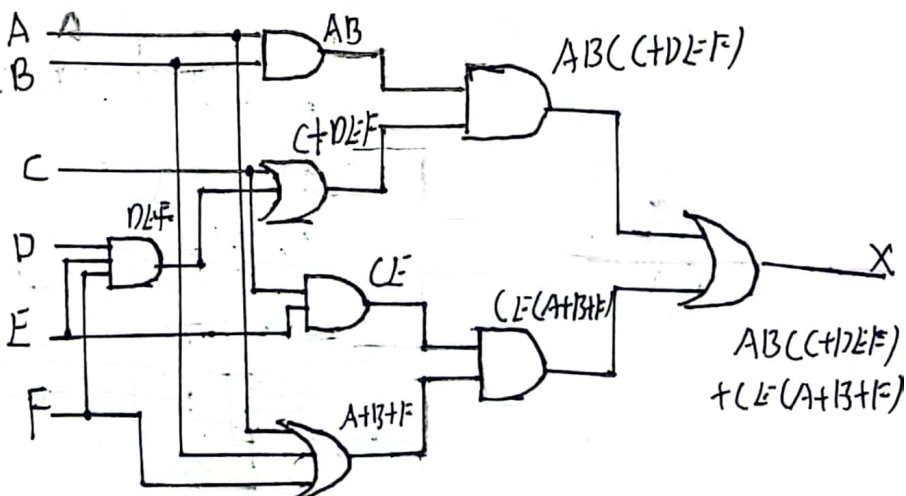
$$= (\bar{A} \cdot B \cdot C) \cdot CD$$

$$= \bar{A} \cdot B \cdot C \cdot D$$

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

5-10 解: (a) $X = \bar{A} \cdot B$

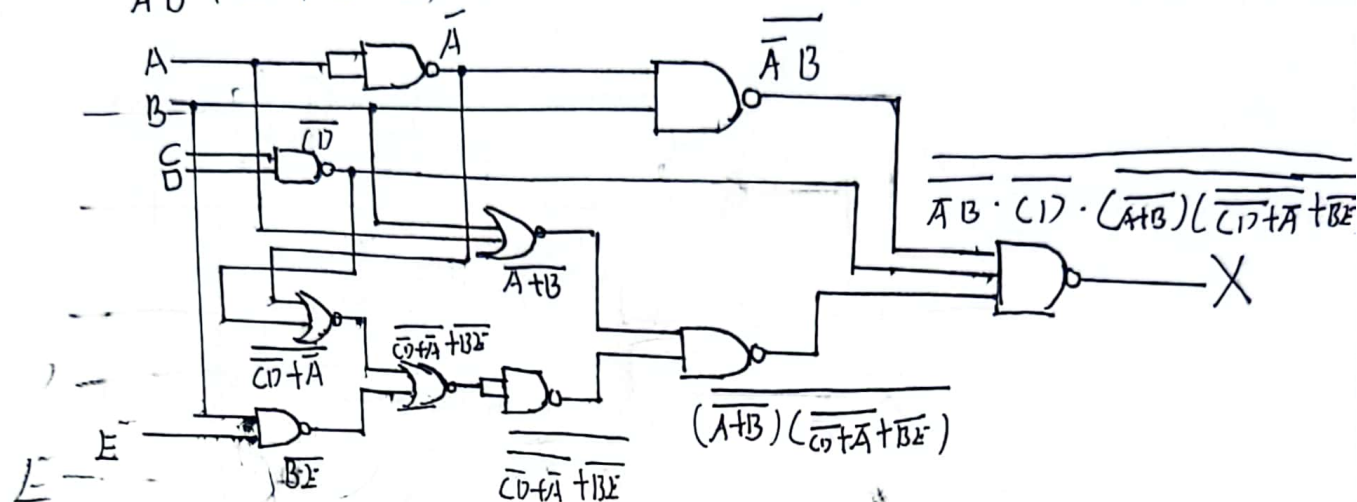
$$AB(C+D)EF + CE(A+B+F)$$



5-12. (a) 解:

$$X = \bar{A}B + CD + (A+B)(ACD + BE)$$

$$= \bar{A}B + CD + (A+B)(\bar{A} + \bar{C}D + BE) = [\bar{A}B \cdot CD \cdot (A+B)(\bar{A} + \bar{C}D + BE)]$$



5-14 解: 5.9

$$X = \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + ABCD$$

$$= \bar{A}\bar{B}C(\bar{D} + D) + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}(\bar{D} + D) + A\bar{B}C(\bar{D} + D) + ABCD$$

$$= \bar{A}\bar{B}C + \bar{A}C\bar{D} \cdot B + A\bar{C}D \cdot B + A\bar{B}\bar{C} + A\bar{B}C = \bar{A}\bar{B}C + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}C + ABCD$$

$$= \bar{A}\bar{B}C + B + A\bar{B}$$

$$= \bar{A}\bar{B}C + B + A\bar{B}C + A\bar{B}$$

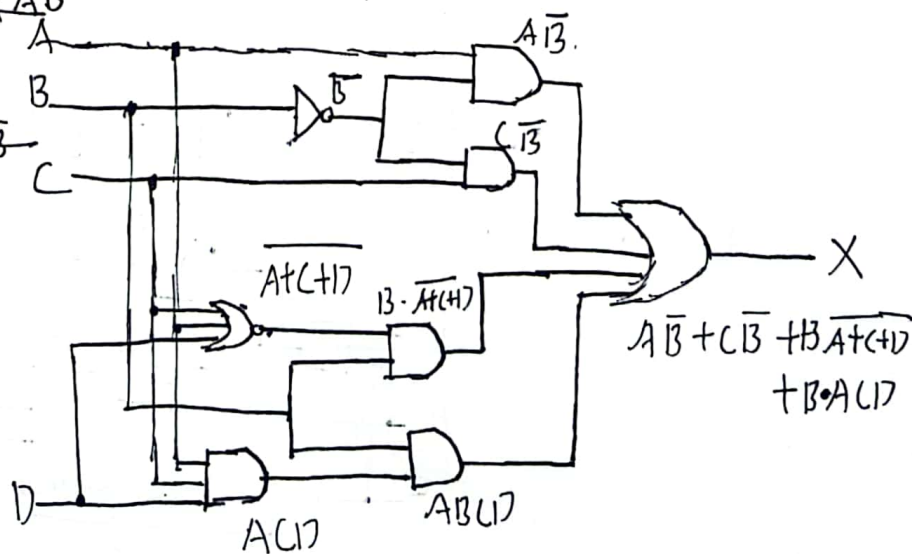
$$= \bar{B}C + B + A\bar{B}$$

$$= \bar{B}C + BC + B + A\bar{B}$$

$$= C + B + A\bar{B}$$

$$= C + B + A + A\bar{B}$$

$$= C + B + A$$



5-20: (a)

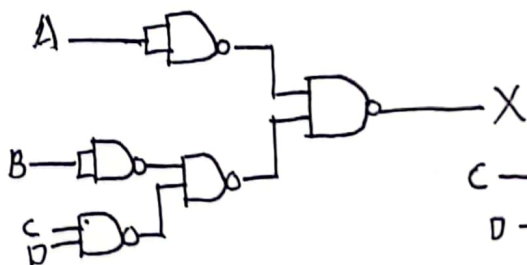
$$\begin{aligned} \text{由5-2可知 } X &= \bar{A} + AC \\ &= \bar{A} + C = \overline{A\bar{C}} \end{aligned}$$

∴ 电路为



(b) 由5-2可知

$$\begin{aligned} X &= A + \bar{B} \cdot \bar{C}D \\ &= A + \overline{B+C} \\ &= \overline{\bar{A} \cdot (B+C)} \\ &= \overline{\bar{A} \cdot \bar{B} \cdot \bar{C}D} \end{aligned}$$

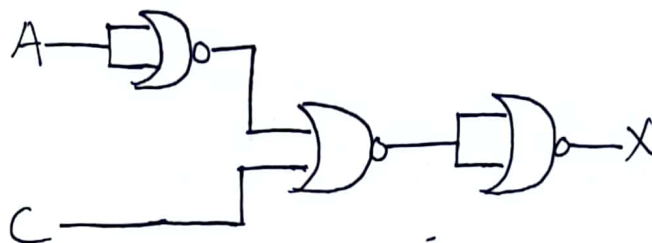


5-22

(a) 由5-2可知

$$X = \bar{A} + C = \overline{\overline{\bar{A} + C}}$$

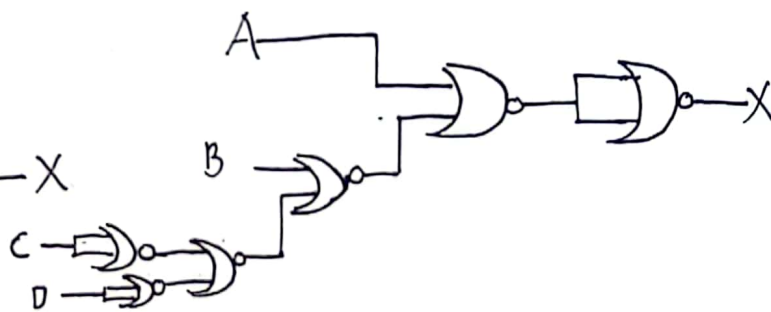
∴ 电路为



(b) 由5-2可知

$$\begin{aligned} X &= A + \bar{B} \cdot \bar{C}D \\ &= A + \overline{B+C} \\ &= \overline{\bar{A} \cdot \bar{B} \cdot \bar{C}D} \end{aligned}$$

电路为



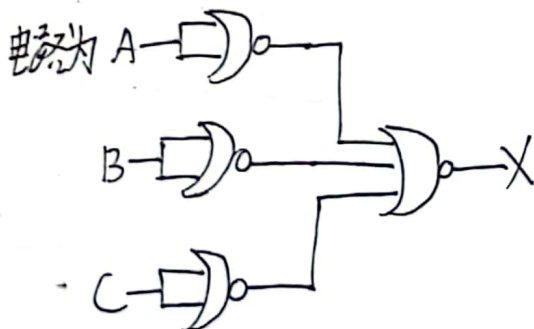
5-24解:

(a) $X = \overline{A+B} \cdot C$

$$= \overline{A+B} \cdot C$$

$$= \overline{\overline{A+B}} + \bar{C}$$

$$= \overline{A+B+C}$$

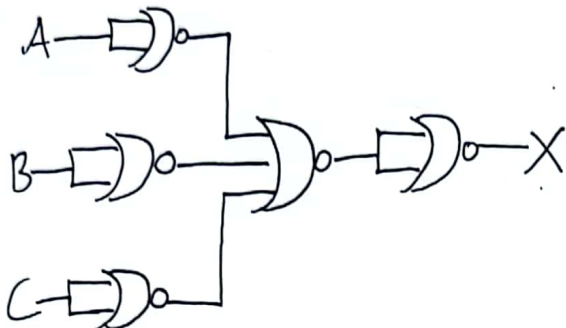


$$(b) X = \overline{ABC}$$

$$= \overline{A+B+C}$$

$$= \overline{\overline{\overline{A+B+C}}}$$

电路:



5-30解:

真值表为

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

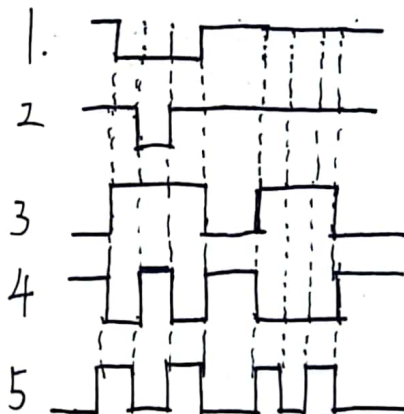
$$5-32 | : \overline{A+B} = \overline{A \cdot B} = X_1$$

$$2: \overline{D+E} = \overline{DE} = X_2$$

$$3: \overline{AB \cdot C} = AB + \overline{C} = \overline{X_1} + \overline{C} = X_3$$

$$4: \overline{AB \cdot C + DE} = \overline{X_3} + \overline{X_2} = X_4$$

$$5: \overline{\overline{AB \cdot C + DE} + F} = \overline{X_4 + F}$$



$$\therefore X = A\overline{B}\overline{C} + A\overline{B}C + ABC$$

$$= A\overline{B}\overline{C} + AC$$

$$= A\overline{B}\overline{C} + A\overline{B}C + AC$$

$$= A\overline{B} + AC$$

